Homework 1 — Comparing AI Tools in Biostatistics: A Critical Evaluation

Course: PM599: AI as a Tool for Biostatistics **Due:** Before Week 2 lecture — May 28, 2026, 12:59 PM **Submission:** Brightspace (PDF or Word document) **Weight:** 10% of final grade (10 points total)

Overview

In this assignment, you will practice using multiple LLM platforms for a biostatistics task and critically evaluate the quality of their outputs. The goal is to develop a habit of thoughtful, skeptical engagement with AI — not just accepting what is produced, but interrogating it.

Reminder: Before starting, make sure you have completed the privacy setup and personalization steps covered in the lab session (May 26). Do not submit sensitive or identifiable data to any AI platform.

Step 1 — Choose a Task

Select **one** task from the list below:

Option A: Explain the assumptions of linear regression to a non-statistician.

Option B: Interpret the following result in plain language:

> In a logistic regression model predicting 30-day hospital readmission, the odds ratio for the variable "diabetes" was 1.84 (95% CI: 1.21–2.79, p = 0.004), adjusted for age, sex, and insurance status.

Option C: Summarize the following abstract and identify one potential limitation:

> **Background:** Large language models (LLMs) have demonstrated impressive performance on medical licensing examinations, but their utility for clinical decision-making remains unclear. **Methods:** We prospectively enrolled 500 patients admitted to a tertiary care hospital and compared attending physician diagnoses to diagnoses generated by GPT-4 using structured admission notes. A panel of three senior physicians adjudicated disagreements. **Results:** GPT-4 agreed with attending physician diagnoses in 73% of cases (95% CI: 69–77%). Agreement was lower for rare conditions (52%) and cases involving social determinants of health (61%). **Conclusions:** GPT-4 shows moderate diagnostic agreement with experienced clinicians but has notable gaps in complex and socially complex cases.

Step 2 — Write and Submit a Prompt

Write a clear, specific prompt for your chosen task and submit it to **at least two** of the following platforms:

- ChatGPT (chatgpt.com — use your USC account for GPT-5.5 Instant access)
- Claude (claude.ai)
- Gemini (gemini.google.com)

Document for each platform:

1. The exact prompt you used (copy-paste verbatim)
2. The full response (copy-paste or screenshot)
3. The model version shown (e.g., GPT-5.5, Claude Opus 4.7, Gemini 3.1 Pro)

Step 3 — Try One Simple Improvement

Make **one** change to your original prompt to improve the response. For example:

- Specify your background (e.g., "I am a biostatistics graduate student")
- Ask for a specific format (e.g., "Answer in three bullet points")
- Specify the audience (e.g., "Explain as if to a public health practitioner with no statistics background")

Re-run the improved prompt on the same platforms and document the responses.

Note: We will cover prompt engineering in depth in Week 2. For now, just try one small improvement and observe what happens.

Step 4 — Written Reflection (~400–500 words)

Write a reflection addressing the following questions. A coherent narrative is fine.

Comparison:

- How did the two platforms differ? Consider accuracy, clarity, and completeness.
- Did either platform produce anything you were skeptical of? How would you verify it?

Your prompt change:

- What happened when you modified your prompt? Was the response noticeably different?

Critical reflection:

- Where did AI help, and where did it fall short for this task?
- Would you use AI assistance like this in your own research? What safeguards would you apply?

Ethics (2–3 sentences):

- Based on the journal policies discussed in Week 1, what would you need to disclose if you used AI assistance while drafting a methods section for submission to *NEJM AI* or *Bioinformatics*?

Submission Format

Your submission should include, in order:

1. **Task selected** (A, B, or C)
2. **Original prompt** and full responses from ≥2 platforms (platform and model noted)
3. **Revised prompt**, explanation of change, and responses
4. **Written reflection** (~400–500 words)

Submit as a single PDF or Word file to Brightspace by May 28, 2026 at 12:59 PM.

Grading Rubric (10 points)

Criterion	Points	Description
Prompt documentation	2	Original and revised prompts complete, verbatim, platforms clearly labeled
Output comparison	3	Substantive comparison across ≥2 platforms with specific observations
Critical reflection	3	Errors identified, limits articulated, ethics question addressed accurately
Prompt revision	2	Change is purposeful; effect on response noted and discussed

There are no "wrong" AI responses. Your analysis is what is graded.

Academic Integrity Note

Using AI tools to complete this assignment is expected and encouraged — that is the point of the exercise. However, your written reflection must represent your own thinking. If you use an AI tool to help draft the reflection itself, you must disclose this in your submission, following the same standards discussed in Week 1 (NEJM AI and Oxford Bioinformatics policies). Undisclosed AI use in the reflection will be treated as an academic integrity violation per the USC Student Handbook.